Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Directional vectors vs quantum state vectors

Files.

- File: `briefs/SuperpositionAndLinearity/SuperpositionAndLinearity.lean`
- Library: `SuperpositionAndLinearity`
- Imports: `ForPhyslib.QuantumMechanics.Qubit.Bloch`
- New file `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean` (imports `Mathlib.Analysis.InnerProductSpace.PiL2`, `Mathlib.LinearAlgebra.Matrix.DotProduct` and physlib's `Physlib.Relativity.PauliMatrices.Basic`): the Bloch vector of a vector of $\mathbb{C}^2$, the Born probability in terms of Bloch vectors, rays and orthogonality.

Setting. $H = \mathbb{C}^2$ (`EuclideanSpace ℂ (Fin 2)`), with $|z^+\rangle = e_0$, $|z^-\rangle = e_1$. The direction and the state vector are

$$v(\theta,\varphi) = (\cos\varphi\sin\theta,\ \sin\varphi\sin\theta,\ \cos\theta), \qquad \psi(\theta,\varphi) = \cos\tfrac{\theta}{2}\, e^{-i\varphi/2}\,|z^+\rangle + \sin\tfrac{\theta}{2}\, e^{i\varphi/2}\,|z^-\rangle\,,$$

(`direction`, `stateVector`). The Bloch vector of $\psi = \psi_0 e_0 + \psi_1 e_1$ is `blochVector` $\psi = (2\,\mathrm{Re}\,\bar{\psi}_0\psi_1,\ \ 2\,\mathrm{Im}\,\bar{\psi}_0\psi_1,\ \ |\psi_0|^2 - |\psi_1|^2)$, the vector of Pauli expectations $\langle\psi|\sigma_i|\psi\rangle$ (`blochVector_apply`). It is not normalized; its length is $\|\psi\|^2$.

Results.

Brief	Lean
eq. (1), $(v^x)^2 + (v^y)^2 + (v^z)^2 = 1$	`direction_dotProduct_self`
eqs. (2), (3), $p(z^+) = \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$	`prob_zp`
eqs. (2), (3), $p(z^-) = \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2}$	`prob_zm`
eq. (3), $\langle\psi|\psi\rangle = 1$	`norm_stateVector`
abstract: $\psi(\theta,\varphi)$ is the state for the direction $v(\theta,\varphi)$	`blochVector_stateVector`: the Bloch vector of $\psi(\theta,\varphi)$ is $v(\theta,\varphi)$
Sec. 2, "linearity is in terms of probability"	`born_stateVector`: $\lvert\langle\psi(\theta',\varphi')\|\psi(\theta,\varphi)\rangle\rvert^2 = (1 + v'\cdot v)/2$; eq. (2) is the case $v' = e_z$ (`stateVector_zero_zero`, `direction_zero_zero`)
Sec. 2, $\theta + 2\pi$: negative components, same direction	`stateVector_add_two_pi`, `direction_add_two_pi`
Sec. 1, states are rays, not vectors	`exists_eq_smul_stateVector_iff`: two state vectors span the same ray if and only if they have the same direction
Sec. 2, the outcomes are opposite directions	`inner_stateVector_eq_zero_iff`: two state vectors are orthogonal if and only if their directions are opposite
abstract: every qubit state is of the form (3)	`exists_eq_smul_stateVector`: every unit vector is a phase times some $\psi(\theta,\varphi)$ with $0 \le \theta \le \pi$
(completeness)	`exists_direction_eq`: every unit vector of $\mathbb{R}^3$ is some $v(\theta,\varphi)$
In `ForPhyslib/QuantumMechanics/Qubit/Bloch.lean`	
$\lvert\psi\rangle\langle\psi\rvert = (\|\psi\|^2 + a\cdot\sigma)/2$	`vecMulVec_eq`
Born probability	`norm_inner_sq`: $\lvert\langle\chi\|\psi\rangle\rvert^2 = (\|\chi\|^2\|\psi\|^2 + a\cdot b)/2$
length of the Bloch vector	`blochVector_dotProduct_self`: $a\cdot a = \|\psi\|^4$
rays	`exists_eq_smul_iff`: unit vectors span the same ray iff their Bloch vectors agree
orthogonality	`inner_eq_zero_iff`: unit vectors are orthogonal iff their Bloch vectors are opposite

Differences from the brief.

- The brief relates the state vector to the direction through the probabilities along z only. The formalization uses the Bloch vector, which fixes the ray completely, so that "the state vector for a qubit, which also corresponds to a direction in physical space" becomes a theorem in

both directions (`blochVector_stateVector`, `exists_eq_smul_stateVector_iff`, `exists_eq_smul_stateVector`, `exists_direction_eq`).

- The brief says the probability "must be proportional to the component in that direction". It is proportional to the length $\overline{Az^-} = 1 + \cos\theta$ in Fig. 2, that is, affine in the component: $p(z^+) = (1 + v \cdot e_z)/2$. The formalization states the affine law for every pair of directions (`born_stateVector`).

- The bijection between rays of $\mathbb{C}^2$ and the unit sphere is proved pointwise (injectivity and surjectivity of the map from state vectors to directions); it is not stated as an equivalence or homeomorphism of the projective line with the sphere.

Problems found in the brief.

- The abstract and eq. (3) write the state as $\cos\frac{\theta}{2}e^{-i\varphi/2}\,|z^+\rangle + \sin\frac{\theta}{2}e^{i\varphi/2}\,|z^+\rangle$. The second ket is $|z^-\rangle$, as the computation of $p(z^-)$ in eq. (3) shows. With $|z^-\rangle$ the signs of the phases are the right ones: the Bloch vector is $v(\theta, \varphi)$ and not its mirror image $v(\theta, -\varphi)$.